

Resolution Geometry of Observable Systems

Observable Quotients, Admissible Distinguishability, Transport, and Entropy Decomposition

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Abstract

We give an axiomatic and modular characterization of Resolution Geometry (RG), understood as the geometry of distinguishability induced by a declared observation protocol. The primitive input is a complete statistical experiment together with an explicit admissibility protocol. Observational extensionality forces equality-of-law equivalence and the universal observational quotient. A local metric module, grounded in sufficient-Markov invariance of information geometry, selects Fisher–Rao distinguishability up to normalization in the regular finite classical setting. Given a protocol metric G and threshold τ , four spectral requirements— G -self-adjointness, idempotence, commutation with the information operator $G^{-1}F$, and threshold consistency—uniquely characterize the resolved projector

$$P_\tau = \mathbf{1}_{(\tau, \infty)}(G^{-1}F).$$

Naturality characterizes its transport under experiment isomorphisms, while identity and cocycle laws characterize gluing into observable atlases. Information monotonicity under parameter-independent garblings and regularity off the discriminant complete the operational axiom core. We prove quotient factorization, representative independence, latent-realization underdetermination, local sector stability, and an entropy decomposition

$$S_G/k_B = H(Q) + H(X \mid Q).$$

Under conditional equiprobability, the fiber term is the expected Boltzmann multiplicity entropy. Computational attacks verify the characterized projector, exhaust all binary spectral selectors in a seven-mode model, test axiom independence, covariance, product composition, quotient universality, continuous-time hidden augmentation, gluing, and the entropy closure on an exactly enumerated $N = 16$ Ising ring. The result characterizes RG relative to the declared experiment and admissibility protocol; it does not eliminate latent states, determine the protocol from nothing, or reduce physics to information.

Contents

1	Introduction	2
2	Primitive object and operational axioms	3
3	Module I: quotient representation	4
4	Module II: characterization of observable distinguishability	4
5	Module III: characterization of the resolved projector	5
6	Module IV: natural transport and gluing	6

7	Modular Characterization Theorem	6
8	Consequences	7
8.1	Latent-realization underdetermination	7
8.2	Observed paths, memory, and holonomy	7
8.3	Information monotonicity	7
9	Entropy as a quotient corollary	7
10	Computational characterization attacks	8
10.1	Axiom-core necessity	8
10.2	Resolved-projector characterization attack	8
10.3	Covariance, product composition, and stability	9
10.4	Metric boundary	9
10.5	Entropy and latent-realization attacks	9
11	Characterized structure and novelty	9
12	Non-claims and falsification criteria	10
13	Conclusion	10

1 Introduction

A mathematical description is observable only relative to an experiment. If two latent states induce the same complete observation law, no statement derived solely from that experiment can distinguish them. The deterministic identity

$$A(x + n) = A(x), \quad n \in \ker A, \quad (1)$$

is merely the linear special case. The general object is the equality-of-law quotient of a statistical experiment.

Resolution Geometry is therefore not postulated as a geometry of spacetime, probability, or latent reality. It is the geometry of observational equivalence classes equipped with the structures required to compare, resolve, transport, and infer between them. Its logical order is

$$\begin{aligned} & \text{declared experiment} \longrightarrow \text{observational quotient} \\ & \longrightarrow \text{distinguishability metric} \longrightarrow \text{admissible sectors and transport.} \end{aligned} \quad (2)$$

Entropy, memory, holonomy, and physical applications are downstream constructions.

The quotient itself is classical. So are Fisher information, spectral projectors, Markov monotonicity, cocycle gluing, and Shannon’s chain rule. The proposed contribution is a modular characterization that composes these established structures into one protocol-explicit observable geometry, states the weakest assumptions needed at each stage, and separates what is forced from what remains a declared experimental convention.

Central characterization. For a regular finite classical experiment and a fixed admissibility protocol, RG is characterized by four modules:

1. observational extensionality and quotient universality;
2. a Fisher–Rao metric characterization under local tensoriality, continuity, normalization, and sufficient-Markov invariance;

3. a unique spectral resolved projector under four projector axioms;
4. natural transport and cocycle gluing.

The resulting local object is

$$\boxed{\text{RG}_\tau(\mathcal{E}, \Gamma) = (Q_\mathcal{E}, g_\mathbb{F}, P_\tau, \{T_{ij}\}, \Gamma).} \quad (3)$$

Uniqueness is relative to the bundled experiment and protocol and up to natural isomorphism and metric normalization.

2 Primitive object and operational axioms

Definition 2.1 (Declared admissible experiment). *A Resolution Geometry problem begins with*

$$\mathfrak{E} = (\mathcal{E}, \Gamma), \quad (4)$$

where

$$\mathcal{E} = (\Theta, \mathcal{O}, \{P_\theta\}_{\theta \in \Theta}) \quad (5)$$

is a complete statistical experiment and

$$\Gamma = (G, \tau, \mathcal{N}, \mathcal{S}, \mathcal{C}, \mathcal{T}) \quad (6)$$

is an explicit admissibility protocol. Here G is a comparison or scale metric, τ a threshold family, $\mathcal{N}$ a nuisance/gauge convention, $\mathcal{S}$ a stability protocol, $\mathcal{C}$ conditioning and coarse-graining rules, and $\mathcal{T}$ transport and gluing data.

The declared object is primitive. It prevents an observable construction from silently using inaccessible latent labels, undeclared priors, regularizers, or external physical assumptions.

Axiom 2.2 (E1: Observational extensionality). *For every experiment-internal construction $\Phi_\mathfrak{E}$,*

$$P_\theta = P_{\theta'} \implies \Phi_\mathfrak{E}(\theta) = \Phi_\mathfrak{E}(\theta'). \quad (7)$$

Axiom 2.3 (E2: Naturality). *Observable constructions commute with isomorphisms of complete experiments and with covariant transport of the bundled protocol. If $f : \mathfrak{E} \rightarrow \mathfrak{E}'$ is an experiment isomorphism, then the corresponding observable construction satisfies the appropriate naturality square.*

Axiom 2.4 (E3: Information order). *A parameter-independent garbling cannot increase distinguishability. In a local metric realization, if $\mathfrak{E}'$ is a garbling of $\mathfrak{E}$, then*

$$g_{\mathfrak{E}'} \preceq g_\mathfrak{E}. \quad (8)$$

Axiom 2.5 (E4: Regularity off the discriminant). *Observable structures vary continuously and retain locally constant discrete invariants away from an explicitly declared discriminant set containing threshold crossings, rank changes, conditioning boundaries, nuisance-rank changes, stability failures, and transport singularities.*

Axiom 2.6 (E5: Composition and gluing). *Local comparison maps preserve identities and composition. On overlapping observable charts,*

$$T_{ii} = \text{id}, \quad T_{jk}T_{ij} = T_{ik}. \quad (9)$$

Remark 2.7 (Independence of roles). E1 excludes inaccessible representative dependence; E2 excludes coordinate-labelled constructions; E3 excludes anti-informational orderings; E4 excludes unstable claims away from genuine transition sets; and E5 excludes path-dependent global comparison. None of E2–E5 follows from quotient factorization alone.

3 Module I: quotient representation

Definition 3.1 (Observational equivalence). *Define*

$$\theta \sim_{\mathcal{E}} \theta' \iff P_{\theta} = P_{\theta'}. \quad (10)$$

Let

$$Q_{\mathcal{E}} = \Theta / \sim_{\mathcal{E}} \quad (11)$$

with canonical projection $\pi_{\mathcal{E}} : \Theta \rightarrow Q_{\mathcal{E}}$.

Theorem 3.2 (Quotient Representation Theorem). *Under E1, every experiment-internal construction $\Phi : \Theta \rightarrow Z$ factors uniquely through the observational quotient. There exists a unique $\tilde{\Phi} : Q_{\mathcal{E}} \rightarrow Z$ such that*

$$\Phi = \tilde{\Phi} \circ \pi_{\mathcal{E}}. \quad (12)$$

Proof. Set $\tilde{\Phi}([\theta]) = \Phi(\theta)$. E1 makes this independent of the representative. Existence follows immediately. If $\Psi \circ \pi_{\mathcal{E}} = \Phi$, surjectivity of $\pi_{\mathcal{E}}$ gives $\Psi([\theta]) = \Phi(\theta) = \tilde{\Phi}([\theta])$, proving uniqueness. $\square$

Corollary 3.3 (Representative independence). *No experiment-internal conclusion may depend on the representative selected inside $[\theta] \in Q_{\mathcal{E}}$.*

Corollary 3.4 (Kernel quotient). *For a deterministic linear map $A : X \rightarrow O$,*

$$x \sim_A x' \iff x - x' \in \ker A, \quad (13)$$

and $Q_A \cong X / \ker A$.

Corollary 3.5 (Universal core). *The observational quotient is universal among maps constant on equality-of-law classes.*

Non-claim 3.6 (No ontological elimination). The theorem establishes operational non-identifiability relative to $\mathcal{E}$. It does not imply that latent representatives do not exist or that another experiment cannot distinguish them.

4 Module II: characterization of observable distinguishability

The quotient does not by itself select a metric. A metric characterization requires additional local assumptions.

Assumption 4.1 (Local metric module). *On regular finite classical statistical models, local distinguishability is represented by a family of symmetric positive-semidefinite bilinear forms satisfying:*

1. *local smooth tensoriality;*
2. *continuity;*
3. *invariance under sufficient Markov embeddings, equivalently the appropriate monotonicity/equality requirement for lossless statistical transformations;*
4. *a fixed positive normalization convention.*

Theorem 4.2 (Fisher–Rao metric characterization). *Under the local metric module, the observable distinguishability metric is the Fisher–Rao metric up to the fixed normalization.*

Grounding. This is the classical information-geometric characterization supplied by a Čencov-type theorem in the regular finite classical setting. RG uses this result as a module; quotient existence alone does not imply Fisher geometry. $\square$

For a local Gaussian model with mean Jacobian J and covariance $C \succ 0$,

$$F = J^\top C^{-1} J. \quad (14)$$

A protocol metric $G \succ 0$ supplies the dimensionless comparison structure. The local information operator is

$$L = G^{-1} F, \quad (15)$$

and its generalized eigenproblem is

$$F v_i = \lambda_i G v_i. \quad (16)$$

Remark 4.3 (Scale and protocol). The Fisher metric is characterized only up to scale before normalization. The comparison metric G and threshold τ are not generated by quotient logic. They belong to the declared protocol or require a separate physical/statistical characterization.

5 Module III: characterization of the resolved projector

Axiom 5.1 (S1: Idempotence). *A hard resolved/null selector is a projector:*

$$P_\tau^2 = P_\tau. \quad (17)$$

Axiom 5.2 (S2: Metric orthogonality). *The selector is G -self-adjoint:*

$$P_\tau^\top G = G P_\tau. \quad (18)$$

Axiom 5.3 (S3: Information compatibility). *The selector commutes with the information operator:*

$$P_\tau L = L P_\tau. \quad (19)$$

Axiom 5.4 (S4: Threshold consistency). *For every generalized information eigenvector,*

$$P_\tau v_i = \begin{cases} v_i, & \lambda_i > \tau, \\ 0, & \lambda_i \leq \tau. \end{cases} \quad (20)$$

Axiom 5.5 (S5: Product composability). *For independent direct-sum experiments with a common threshold convention,*

$$P_\tau(L_1 \oplus L_2) = P_\tau(L_1) \oplus P_\tau(L_2). \quad (21)$$

Theorem 5.6 (Resolved Projector Characterization). *Given (F, G, τ) with $G \succ 0$ and $\tau \notin \text{spec}(L)$, the unique operator satisfying S1–S4 is*

$$\boxed{P_\tau = \mathbf{1}_{(\tau, \infty)}(L)}. \quad (22)$$

It also satisfies S5.

Proof. Because L is self-adjoint in the G inner product, it admits a G -orthonormal generalized eigenbasis. S3 makes every eigenspace invariant under P_τ . S1 and S2 restrict P_τ on each eigenspace to an orthogonal projector. S4 fixes that action to the identity above threshold and zero at or below threshold. Hence (22) is unique. Functional calculus on direct sums gives S5. $\square$

Definition 5.7 (Resolved, null, and admissible sectors). *Define*

$$R_\tau = \text{im } P_\tau, \quad N_\tau = \ker P_\tau. \quad (23)$$

The final sector $W_{\text{obs|adm}} \subseteq R_\tau$ is obtained after nuisance closure, stability, conditioning, coarse-graining compatibility, and transport gates in Γ . Its effective complement is N_{adm} .

Remark 5.8 (Soft resolution). If S1 is removed, smooth spectral filters become admissible. They define graded resolution but not a hard resolved/null decomposition. Hard sectors therefore require idempotence; the broader RG framework may also study soft filters explicitly.

6 Module IV: natural transport and gluing

Theorem 6.1 (Naturality of the resolved projector). *Let an experiment isomorphism be represented locally by invertible parameter and observation transformations*

$$\theta' = S\theta, \quad y' = Uy, \quad (24)$$

with

$$F' = S^{-\top} F S^{-1}, \quad G' = S^{-\top} G S^{-1}. \quad (25)$$

Then the generalized spectrum is invariant and

$$\boxed{P'_\tau = S P_\tau S^{-1}}. \quad (26)$$

Proof. If $Fv = \lambda Gv$, then $v' = Sv$ satisfies $F'v' = \lambda G'v'$. Functional calculus therefore transports the threshold projector by conjugation. $\square$

Definition 6.2 (Observable atlas). *An observable atlas consists of local quotient charts, local admissible sectors, and bounded transports T_{ij} on overlaps satisfying E5.*

Theorem 6.3 (Transport and gluing characterization). *Natural local sector transports define a consistent global observable atlas if and only if identity and cocycle compatibility hold on all overlaps.*

Proof. Identity and cocycle closure make direct and composite identifications agree. If the cocycle fails, the same local element acquires path-dependent images, obstructing a global identification. $\square$

7 Modular Characterization Theorem

Theorem 7.1 (Modular Characterization of Resolution Geometry). *Let $(\mathcal{E}, \Gamma)$ be a regular finite-dimensional classical admissible experiment. Assume E1–E5, the local metric module, and S1–S5. Assume also that τ is separated from the local generalized spectrum and that declared nuisance and stability gates are satisfied. Then:*

1. *every experiment-internal construction factors uniquely through $Q_\mathcal{E}$;*
2. *the normalized local observable metric is Fisher–Rao;*
3. *the unique hard resolved selector is $P_\tau = \mathbf{1}_{(\tau, \infty)}(G^{-1}F)$;*
4. *(R_τ, N_τ) is natural under experiment isomorphisms;*
5. *compatible local sectors glue under the transport cocycle;*

6. sector rank is locally constant away from the discriminant.

Consequently,

$$\boxed{\text{RG}_\tau(\mathcal{E}, \Gamma) = (Q_\mathcal{E}, g_\mathcal{F}, P_\tau, \{T_{ij}\}, \Gamma)} \quad (27)$$

is characterized up to natural isomorphism and the declared metric normalization, relative to the bundled protocol.

Proof. Items (i)–(v) follow respectively from the Quotient Representation Theorem, the Fisher–Rao metric characterization, the Resolved Projector Characterization, projector naturality, and the gluing theorem. Item (vi) follows from E4 and continuity of symmetric-definite generalized eigenvalues away from threshold crossings and other gate boundaries. $\square$

Non-claim 7.2 (Scope of characterization). The theorem does not derive τ , G , nuisance conventions, stability classes, or physical interpretation from no assumptions. It characterizes the RG object once the experiment and admissibility protocol are declared and the stated modular axioms hold.

8 Consequences

8.1 Latent-realization underdetermination

Theorem 8.1 (Latent-Realization Underdetermination). *If two latent models induce isomorphic quotient experiments and the same transported protocol, their characterized RG objects are naturally isomorphic. Experiment-internal inference cannot uniquely determine latent dimension, redundant coordinates, hidden dynamics annihilated by observation, or a representative inside an observational class.*

Proof. Quotient universality identifies the experiment-internal domain. The remaining characterized modules are natural under the quotient-experiment isomorphism. $\square$

8.2 Observed paths, memory, and holonomy

For stochastic experiments, the observation space may be a path space. Every memory functional, path integral, or holonomy determined by the observed path law factors through the path-law quotient. A hidden-path functional is not experiment-internal unless the experiment is enlarged or a prior/model assumption is declared.

8.3 Information monotonicity

For a parameter-independent garbling, the score of the processed observation is the conditional expectation of the original score. Conditional Jensen therefore yields

$$F_{\text{processed}} \preceq F_{\text{original}}. \quad (28)$$

Thus E3 is realized by Fisher geometry in regular statistical models.

9 Entropy as a quotient corollary

Entropy is not an axiom of RG. It is a downstream construction once a probability measure is placed on a microstate space and pushed through an observational quotient.

Theorem 9.1 (Observable-Quotient Entropy Decomposition). *Let X be a discrete microstate variable and $Q = \pi(X)$ a deterministic observational quotient. Then*

$$H(X) = H(Q) + H(X | Q). \quad (29)$$

If $S_G = k_B H(X)$, then

$$\boxed{S_G = k_B H(Q) + k_B H(X | Q)}. \quad (30)$$

Proof. Since Q is a deterministic function of X , $H(X, Q) = H(X)$. Apply the Shannon chain rule. $\square$

Corollary 9.2 (Boltzmann entropy in quotient fibers). *If $X | Q = q$ is uniform on a fiber of multiplicity $\Omega(q)$, then*

$$H(X | Q = q) = \log \Omega(q) \quad (31)$$

and

$$S_G = k_B H(Q) + \mathbb{E}_Q[k_B \log \Omega(Q)]. \quad (32)$$

Corollary 9.3 (Equality boundary). *The quotient Shannon entropy equals total Gibbs entropy in units of k_B if and only if*

$$H(X | Q) = 0, \quad (33)$$

equivalently, π is injective on $\text{supp } p$.

Corollary 9.4 (Resolution redistribution). *If Q_f refines Q_c , then*

$$H(Q_f) \geq H(Q_c), \quad H(X | Q_f) \leq H(X | Q_c), \quad (34)$$

while $H(X)$ remains fixed. Increased resolution transfers entropy from unresolved fibers to observable classes; it does not create total entropy.

10 Computational characterization attacks

The numerical batteries do not prove the classical quotient, spectral, Fisher, or Shannon theorems. They search for implementation failures, hidden assumptions, and counterexamples to stronger claims.

10.1 Axiom-core necessity

The axiom-core battery constructed alternatives after removing individual operational axioms. A representative-index selector varied within equality-of-law classes. A coordinate-labelled selector violated naturality. The negative Fisher trace was quotient-based and natural but increased under garbling. A hard threshold was stable off the discriminant but changed rank under arbitrarily small boundary perturbations. Bounded pairwise transports with a perturbed triple-overlap map produced path-dependent comparison. These witnesses show that extensionality, naturality, information order, off-discriminant regularity, and gluing do different logical work.

10.2 Resolved-projector characterization attack

A seven-mode generalized-Fisher problem was used to test S1–S5. The canonical projector produced

$$\|P^2 - P\| = 8.29 \times 10^{-16}, \quad (35)$$

$$\|P^\top G - GP\| = 2.80 \times 10^{-16}, \quad (36)$$

$$\|PL - LP\| = 9.53 \times 10^{-15}, \quad (37)$$

with the expected rank four. Exhaustive enumeration of all $2^7 = 128$ binary spectral projectors left exactly one candidate satisfying the threshold mask and all projector axioms:

$$(0, 0, 0, 1, 1, 1, 1). \quad (38)$$

Removing threshold consistency admitted a different orthogonal spectral projector. Removing commutation admitted a random G -orthogonal projector with commutator norm 7.36. Removing idempotence admitted a smooth logistic filter with idempotence error 0.325. Removing self-adjointness admitted an oblique projection with metric-adjoint error 1.64. Thus the projector axioms are independently active.

10.3 Covariance, product composition, and stability

Coordinate scale spans through 10^3 preserved resolved rank four; at the largest span the relative projector naturality error was 3.20×10^{-5} , attributable to numerical conditioning. For an independent direct-sum experiment,

$$\|P_\tau(L_1 \oplus L_2) - P_\tau(L_1) \oplus P_\tau(L_2)\| = 2.87 \times 10^{-15}. \quad (39)$$

With spectral margin 0.2462, perturbations of norm 10^{-9} through 10^{-3} retained rank four and changed the projector continuously.

10.4 Metric boundary

An arbitrary positive-definite covariant metric differed from the Fisher tensor by norm 13.35 while satisfying positivity and transportability. Therefore quotient universality, covariance, regularity, and composition do not select Fisher. The separate Čencov-type metric module is indispensable.

10.5 Entropy and latent-realization attacks

Earlier attacks constructed finite, nonlinear, chaotic, and continuous-time latent realizations with identical quotient experiments and quotient Fisher geometry. An exactly enumerated $N = 16$ Ising ring closed

$$H(X) = H(Q) + H(X | Q) \quad (40)$$

to approximately 10^{-13} or better across magnetization, energy, and joint quotients and across seven inverse temperatures. At conditional equiprobability, $H(X | Q)$ agreed with $\mathbb{E}[\log \Omega(Q)]$ to 1.78×10^{-15} .

11 Characterized structure and novelty

The observational quotient is classical. Fisher–Rao geometry is classical. Spectral projectors, perturbation theory, Markov monotonicity, cocycle gluing, and Shannon entropy are classical. The present claim is not that these components were individually invented by RG.

The contribution is their modular characterization as one observable geometry:

extensionality $\implies Q_\mathcal{E}$, sufficient-Markov metric axioms $\implies g_F$, spectral sector axioms $\implies P_\tau$, naturality and cocycles $\implies$ observable atlas.	(41)
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This identifies both the unavoidable core and the precise choices that remain protocol-relative.

A stronger theorem might characterize the entire protocol or derive a preferred threshold from decision loss, finite-resource constraints, or a physical noise floor. Such a theorem is not established here.

12 Non-claims and falsification criteria

Non-claim 12.1 (No geometry of latent reality). RG is the geometry induced by a declared observation protocol, not a claim that the latent world itself is exhausted by the quotient.

Non-claim 12.2 (No protocol-free uniqueness). The theorem does not choose G , τ , nuisance equivalence, or stability class without additional assumptions.

Non-claim 12.3 (No universal information–physics identity). The equality $S_G = k_B H(Q)$ holds only at $H(X | Q) = 0$. The quotient does not select energy, equilibrium, temperature, or k_B .

Non-claim 12.4 (No unrestricted initiality of $W_{\text{obs|adm}}$). The observational quotient has the universal factorization property. A thresholded admissible sector is characterized relative to (F, G, τ, Γ) , not as an initial object among every conceivable observable functor.

The framework is falsified or requires revision if an experiment-internal quantity varies inside an equality-of-law class; if a correctly transported experiment isomorphism changes the characterized spectrum or projector; if a parameter-independent garbling increases the characterized information tensor; if a claimed stable sector has no positive discriminant margin; if global transports violate cocycle closure; or if an entropy identity omits a nonzero conditional fiber term.

13 Conclusion

The axiomatic foundation of Resolution Geometry is modular rather than monolithic. Observational extensionality forces the quotient and its universal factorization. Established sufficient-Markov metric characterization selects Fisher–Rao geometry in its domain of validity. Spectral axioms uniquely select the hard resolved projector. Naturality and cocycle laws characterize comparison and gluing. Regularity restricts stable claims to strata away from explicit discriminants.

The resulting construction is

$$\boxed{\text{RG}_\tau(\mathcal{E}, \Gamma) = (Q_\mathcal{E}, g_\mathcal{F}, P_\tau, \{T_{ij}\}, \Gamma),} \quad (42)$$

unique up to natural isomorphism and metric normalization relative to the declared protocol. Its strongest operational consequence is

$$\boxed{\text{A latent point is not an experiment-internal observable object unless the experiment separates it.}} \quad (43)$$

This conclusion neither denies latent reality nor derives physics from information. It identifies the canonical domain and characterized geometry of what a declared experiment can support.

Reproducibility

The supporting artifacts are:

- `rg_axiom_core_stress_battery.json`;
- `rg_characterization_theorem_attack.json`;
- `rg_observable_quotient_invariance_battery.json`;
- `rg_admexp_category_lock_battery.json`;
- `rg_final_universality_attack.json`;
- `rg_quotient_entropy_bridge_attack.json`.

The axiom-core and characterization attacks used deterministic seed 20260804; prior quotient and universality attacks used seeds 20260802 and 20260803. The Ising entropy attack was exactly enumerable and used no random seed.

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